

SHARJAH INTL, United Arab Emirates

WMO: 411960

Lat: 25.329N

Lon: 55.517E

Elev: 34

StdP: 100.92

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.8	12.0	-0.4	3.7	29.0	1.9	4.3	28.1	9.4	21.7	8.6	21.5	2.1	120	0.416

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	44.3	23.5	43.1	23.7	42.0	24.1	30.0	36.2	29.4	35.8	28.8	35.3	4.3	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.8	25.4	33.6	28.0	24.2	33.2	27.2	23.0	33.0	100.8	36.1	97.5	35.7	94.5	35.1	33.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.4	6.5	DB	7.6	47.1	1.7	0.7	6.4	47.6	5.5	48.0	4.5	48.4	3.3	48.9	(4)
(5)			WB	5.5	31.3	1.7	0.7	4.3	31.8	3.3	32.2	2.4	32.6	1.2	33.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	19.0	20.4	23.3	27.5	31.9	34.3	36.2	36.0	33.3	29.7	24.8	20.7	(6)	
(7)		DBStd	6.45	1.97	2.66	2.93	2.96	2.25	1.70	1.69	1.48	1.60	1.75	2.14	2.21	(7)	
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
(9)		HDD18.3	28	16	7	1	0	0	0	0	0	0	0	0	4	(9)	
(10)		CDD10.0	6623	278	291	412	526	679	729	813	807	699	611	445	333	(10)	
(11)		CDD18.3	3609	35	66	154	276	421	479	554	549	449	352	195	79	(11)	
(12)		CDH23.3	51335	213	479	1373	3350	6268	7644	9235	9040	6788	4527	1866	553	(12)	
(13)	CDH26.7	33207	24	139	564	1847	4063	5291	6758	6564	4479	2613	762	101	(13)		
(14)	Wind	WSAvg	3.0	2.8	3.1	3.2	3.2	3.4	3.3	3.3	3.1	2.8	2.6	2.6	(14)		
(15)	Precipitation	PrecAvg	84	19	32	23	9	1	0	0	0	0	4	18	(15)		
(16)		PrecMax	167	98	143	125	44	21	0	1	0	9	0	39	146	(16)	
(17)		PrecMin	9	0	0	0	0	0	0	0	0	0	0	0	0	(17)	
(18)		PrecStd	53	26	41	34	13	5	0	0	0	2	0	10	34	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.0	33.1	36.9	40.8	43.8	45.1	46.1	45.9	43.2	39.8	35.1	31.0	(19)	
(20)			MCWB	17.5	17.8	18.6	19.7	21.1	22.2	23.1	22.8	22.8	21.4	19.6	18.9	(20)	
(21)		2%	DB	27.1	30.2	34.1	38.7	42.0	43.2	44.7	44.1	41.7	38.1	33.2	29.0	(21)	
(22)			MCWB	17.2	17.6	18.5	19.6	21.4	22.8	24.1	24.1	23.3	21.8	20.3	18.9	(22)	
(23)		5%	DB	25.8	28.1	32.0	36.9	40.6	42.0	43.0	42.9	40.2	36.9	32.1	27.8	(23)	
(24)			MCWB	17.2	17.5	18.3	19.5	21.5	23.6	24.6	24.5	23.8	22.2	20.7	19.1	(24)	
(25)		10%	DB	24.1	26.1	29.9	35.0	39.0	40.6	41.8	41.2	39.0	35.8	30.9	26.3	(25)	
(26)			MCWB	17.0	17.6	18.2	19.8	21.7	24.0	25.2	25.2	24.4	22.6	20.9	18.5	(26)	
(27)		Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.7	21.2	22.5	24.5	27.4	29.9	31.1	30.7	30.0	28.2	24.7	22.7	(27)
(28)				MCD B	24.9	26.1	28.3	32.6	35.2	36.5	37.4	36.6	35.8	33.3	30.0	27.0	(28)
(29)	2%		WB	19.5	20.2	21.5	23.2	26.2	29.0	30.1	30.0	29.1	27.3	23.7	21.4	(29)	
(30)			MCD B	24.0	25.3	27.1	30.9	33.9	35.5	36.6	35.8	34.6	32.5	29.2	25.9	(30)	
(31)	5%		WB	18.7	19.5	20.8	22.4	25.4	28.2	29.4	29.2	28.6	26.5	23.0	20.6	(31)	
(32)			MCD B	23.4	24.7	26.4	30.1	33.4	34.7	36.2	35.6	34.0	31.8	28.7	25.3	(32)	
(33)	10%		WB	17.9	18.7	20.1	21.7	24.5	27.5	28.8	28.6	27.9	25.8	22.3	19.6	(33)	
(34)			MCD B	22.6	23.9	25.8	29.5	32.9	34.3	35.8	35.4	33.4	31.3	28.1	24.5	(34)	
(35)	Mean Daily Temperature Range		5% DB	MDBR	10.9	11.7	12.1	13.8	14.6	13.9	12.4	12.7	13.3	13.5	12.2	11.4	(35)
(36)				MCDBR	13.1	14.3	15.6	16.7	16.8	15.9	14.2	14.4	14.6	14.7	13.6	13.0	(36)
(37)		MCWBR		5.8	6.0	6.2	6.4	6.4	6.6	6.2	6.1	6.3	6.1	5.7	5.9	(37)	
(38)		5% WB	MCD BR	11.1	11.5	11.8	13.7	14.5	13.4	11.6	12.1	12.7	12.6	12.3	11.4	(38)	
(39)			MCWBR	5.7	5.6	5.4	6.1	6.4	6.1	5.5	5.6	5.6	5.5	5.4	5.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.438	0.497	0.557	0.590	0.586	0.636	0.724	0.686	0.596	0.512	0.453	0.435	(40)		
(41)		taud	2.094	1.922	1.779	1.731	1.739	1.680	1.564	1.634	1.769	1.928	2.082	2.104	(41)		
(42)		Ebn at Noon	799	777	754	741	742	701	641	663	713	753	777	782	(42)		
(43)		Edh at Noon	143	182	220	236	234	247	276	257	220	178	143	136	(43)		
(44)	All-Sky Solar Radiation	RadAvg	4.01	4.80	5.68	6.58	7.38	7.53	7.11	6.70	6.20	5.39	4.35	3.79	(44)		
(45)		RadStd	0.26	0.39	0.39	0.40	0.15	0.14	0.18	0.15	0.09	0.08	0.13	0.25	(45)		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.55	+0.61	N/A	N/A	N/A	N/A	N/A	N/A	+146	+141		
(47)	Regional (14 neighbors)	+0.47	+0.70	N/A	N/A	N/A	N/A	N/A	N/A	+150	+149		

Nomenclature: See separate page